

Ángela Izquierdo García

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Cryptography Engineer | Data Scientist | Data Engineer

Experience

Freelance Developer, Independent Projects / Remote

Mar 2023 - Oct 2025

- Developed multiple independent projects and short-term collaborations while traveling, gaining practical experience in React, Node.js, Express, PostgreSQL, and JavaScript/TypeScript.
- Managed full project lifecycle: design, implementation, testing, and deployment.
- Delivered functional solutions that provided sufficient income to support living expenses during this period.
- Focused on learning, skill development, and practical application of technologies in real-world scenarios.

Self-Directed Trading, Personal Account

2022 – Present

- Designed and tested discretionary and systematic trading approaches focused on futures markets and liquidity-driven behavior.
- Performed data analysis and strategy evaluation using Python and statistical techniques.
- Conducted extensive backtesting and journaling to improve decision-making consistency and risk management.
- Applied disciplined risk management and capital preservation principles.

Apple, Cryptography Engineer | Remote | Cupertino (California, USA)

Jun 2022 - Dec 2022

- Took ownership of core cryptographic algorithms in the WebKit engine, focusing on reliability and performance.
- Refactored legacy crypto modules, achieving up to a **22% improvement in execution speed** and code maintainability.
- Fixed critical bugs in secure communication routines, reducing error rates by **18%** and improving compatibility across platforms.

SciTheWorld (Himitsu Tech), Data Scientist | Madrid, Spain

Jul 2021- Jun 2022

- Developed a scalable pipeline to process **hundreds of 300-page documents daily** for OCR-based NLP tasks.
- Enhanced pre-processing routines and indexing logic, improving pipeline efficiency by approximately **20%**.
- Built internal tools to analyze **liquidity-driven strategies** in futures trading, achieving a discretionary strategy accuracy of **70–80%**.
- Optimized backend services supporting the trading platform, improving API responsiveness and system reliability.
- Contributed to the design and migration of ETL pipelines and databases for **Banco Santander** to AWS.

Skills

Programming Languages	Python, C++, Rust, GO, JavaScript, R, Java, PHP, Ruby
Web Development	HTML/CSS, jQuery, Django
Databases	MongoDB, MySQL, PostgreSQL
Data Science	Data Analysis, Data Cleaning, Machine Learning, Clustering, Predictive Modeling, Statistics, NLP, Data Visualization (Matplotlib, Seaborn)
Cryptography	Post-Quantum Cryptography, Lattice-Based Cryptography, Cryptographic Protocols, Secure Communication, WebKit Crypto Integration
Cybersecurity	Privacy-Preserving Systems, Secure Web Development, Email Validation, Bug Fixing in Critical Systems
Finance & Trading	Experience with futures market trading, developing and testing strategies based on liquidity dynamics
Tools & Platforms	Git, Linux, Docker, GitHub
Soft Skills	Problem-Solving, Analytical Thinking, Communication, Teamwork, Technical Writing

Projects

Chess Game Analytics – Data Science Master Thesis

- Performed advanced analytics on **20,000+ chess games** using Python, Jupyter Notebooks, and open datasets from **Lichess**.
- Applied **machine learning models** (K-Means, DBSCAN, Gradient Boosting, SVM) to predict winners, classify moves, and cluster players.
- Implemented a custom **opening recommendation system** and **move evaluator** using Stockfish and algebraic notation analysis.
- Designed an analytical framework for player evolution, opening efficiency, and decision-making strategies.
- Master's thesis in **Data Science**, Universidad Autónoma de Madrid.

Lattice-Based Cryptography Algorithms (Bachelor Thesis Project)

- Developed and analyzed cryptographic algorithms based on **lattice problems**, with a focus on post-quantum security.
- Implemented core cryptographic primitives, such as **encryption, signature, and hash functions**, using Python.
- Studied **worst-case to average-case** reductions in lattice problems and their implications for cryptographic security.
- Presented a comparative study of lattice-based cryptography vs. classical schemes under potential **quantum attacks**.
- Final project in the **Computer Science and Engineering** degree at UGR.

Education

MSc in Data Science

2022-2024

Universidad Autónoma de Madrid | Madrid, Spain

BSc in Computer Science and Engineering

2016-2021

Escuela Técnica Superior de Ingenierías Informática y de Telecomunicación | Granada, Spain

UGR Top 100 (Shanghai 2024)

BSc in Mathematics

2016-2021

Facultad de Ciencias | Universidad de Granada | Granada, Spain

Certificates

2022	Introduction to Trading, Machine Learning and GCP , Coursera
2022	How Google Does Machine Learning , Coursera

*Online**Online***Languages**

Spanish	Native proficiency
English	Advanced (CAE – C1 by University of Cambridge)
Italian	Advanced (C1 – Centro Linguistico dell'Università Roma Tre)
French	Intermediate (currently studying)
German	Intermediate (currently studying)